Design notes

DIY CNC epoxy-granite mill · October 2026

Fixed-gantry, epoxy-granite desktop mill sized like the Shariff DMC2 Mini (180 x 300 x 140 mm travel), with a 2020/ACM enclosure mounted on the base and a clear lift-up door.

This document explains the design: the frame, the coolant system, the stiffness study and the cost. The repository layout and the rebuild commands are in the top-level `README.md`.

Frame

Five castings that bolt together. Each is a **steel skeleton** of stock flat bar and tube, cut to length, drilled from printed templates and clamped with threaded rod, cast into **epoxy granite**, which damps it and adds stiffness. The steel is the frame: every joint between castings, and every rail, riser and ballscrew support, bolts steel to steel. Nothing is welded, nothing is custom-cut, and nothing is threaded into the epoxy granite (17 M5 inserts hold only the add-ons: the enclosure, the way-cover rollers and the cable-chain mount).

Module	Steel skeleton (all stock)	Joins to
Base, two halves	Per half: a 2x3 tube girder under each Y rail with a tapped 3/8 x 1-1/2 in rail bar bolted on top and a 3/8 x 2-1/2 in end plate at each end; a pedestal under each upright (3/8 x 2-1/2 in seat and bottom bars, a 2x2x1/8 tube spacer round each tie rod); 1-1/2 in cross tubes from girder to girder (at the split and under the Y ballscrew supports, with tapped pads bolted on) and from each girder to its pedestal, each clamped by a 1/2-13 rod	The halves meet end plate to end plate on four dowel pins, clamped by two 1/2-13 draw rods that run inside the girders from a front pocket to a back pocket. The front half is cast against the rear, so the joint fits by construction
Upright (x2), a raked buttress	A 3/8 x 2 in flat-bar foot, two 2x2x1/8 tube columns, a top plate of two 1/2 x 2-1/2 in flat bars (the lower one tapped); O-ring groove under the foot	Its foot bar bears on the base pedestal's seat bar. Three 1/2-13 tie rods (two inside the columns, screwed into the top plate; one from the toe, onto the foot bar) run right through the pedestal to its bottom bar. Two dowels, grouted into the foot at first assembly, locate it
Beam (140 deep x 146 high)	A 1/2 x 5 in face bar (the rail face, tapped) and a 3/8 x 5 in back bar, held apart by three 3 in pipe nipples on studs; a 3/8 x 2-1/2 in seat and cap bar at each end	The seat bars bear on the upright top plates; at each end four 3/8-16 grade B7 studs, screwed into the top plate, clamp cap bar, face and back bars and seat bar down; a dowel (grouted at first assembly) locates it

Nothing is made to order. The rails (including the 600 mm Y pair) and ballscrews are used at the lengths they're sold in; the skeletons are about 54 pieces of stock flat bar and tube (56 kg) cut to length (`docs/1-sourcing/steel/` has the cut list and the drilling templates); the risers and the aluminum table are ordered cut to length (the table is a stock 6061 flat-bar width, 10 in; the X carriage and the Z plate are 5/8 x 6 in steel flat bar, Blanchard ground with the Z risers, the carriage with two 1-1/4 in angles, all bought with the skeletons' steel). Every rod is a grade B7 fully threaded stud bought at a stock length (they come in 1/4 in steps; each is within 3 mm of the length the joint wants, and the tail past its far nut takes the difference), so nothing is sawn at home: the machining is drilling and tapping (from the templates) and the risers. Use plain steel, not galvanized: epoxy bonds poorly to zinc. Wire-brush off rust and loose scale and degrease just before casting; paint only the part of each leg that hangs below the base.

Keeping the epoxy down. About 9.0 kg of epoxy (8.5 % of the 106 kg of castings):

- No mix goes in the tubes (girders, columns, cross tubes, legs): their ends are closed by the plates and bars they're clamped between, or by printed rings and plugs. They're filled with **dry sand** instead (pool filter sand, 6.9 L, 11 kg, tapped down), which damps the tube walls (a loss factor of about 0.15, against 0.02 for epoxy granite and 0.0005 for bare steel) and costs no epoxy. Filling them with epoxy granite would take about 2 kg more epoxy, past the 3 gal kit, and needs a mix without gravel to get round the rods. The rods that come out (the base draw rods and the upright tie rods) run in 1/2 in ID x 5/8 in OD vinyl sleeves, 2.5 m in all, so the sand stays put when they're pulled; the cross rods stay in, so the sand packs round them. Each base tube has a 3/8 in fill hole in the face that's up when the grid lies upside down in the mold, taped over before the pour; the upright columns are filled before the top ring goes in, the legs before their plug, over a printed cup that keeps the leveling foot's stud clear.
- The base's underside pockets follow the stiffness study below, with the steel grid in the ribs between them: two full-depth girders (80 mm wide) under the table rails, a full-depth tie between them across the split, a full-depth rib across the front of each upright foot and another along the split, and a 20 mm top skin almost everywhere else, including over pockets under the upright feet that keep clear of the band under each upright where the pedestal runs. The 70 mm front wall carries the coolant trough. Every pocket is open to the bottom, so the cores (printed shells or foam blocks) pull straight out; a few are shallow steps (40 or 65 mm deep) under the trough band, the draw rods, the knees and outboard of the pedestals (`base_pockets` in `model/build_cnc.py`).

The outside. The base's lower edge is beveled 1-1/2 in in and 2-1/2 in up all round, and the legs are tucked in behind the bevel (65 mm in from the sides and the front, 62 mm from the back, where the pedestal bars are), so the base reads as a 46 mm slab standing on its legs instead of a 110 mm box. It costs nothing. What the bevel takes off is the thin skirt outboard of the pockets, which carried almost nothing. The pockets were pulled in clear of it, so five shallow foam cores are gone. Tucking the legs in shortens the span the base bends over between its feet, which raises every mode by 11-31 % (150 / 292 / 389 Hz against 128 / 256 / 298, the vibration table below) and the stiffness at the tool a little (103.4 / 155.8 / 136.5 N/um against 100.9 / 155.4 / 135.6). The castings are 1 kg lighter. In the mold the bevel is a strip along each wall at the pour line: a 2x3 ripped corner to corner and screwed through the wall from outside, or printed (about 0.7 kg of PETG and 36 hours). The renders show the castings in the gray of the mix as cast (gray aggregate and whiting in clear epoxy). A little black pigment in the resin gives the dark look of commercial castings, and makes it look heavier again.

The uprights' backs slope too: each foot runs 40 mm behind the beam's back and the back face slopes up to the beam, so from the side and the back the uprights read as tapered buttresses instead of blocks. That is the stiff way to add material, low and spread out, where the upright's bending is largest. It adds 0.4 L to each upright (1 kg) and makes the frame stiffer (105.2 / 157.2 / 138.2 N/um against 103.4 / 155.8 / 136.5; modes 150 / 295 / 389 / 398 Hz against 150 / 292 / 389 / 386). The foot stops 10 mm short of the enclosure's back rail. In the mold the back board leans in, like the front board, and the side plates are cut to the new profile. A chamfer under the beam's back edge was tried too: it lost 2 % in Y for little to see, so the beam stays as it is.

Molds and fits. The molds are a kit (`model/moldbox.py`, the mold guide): melamine boards, all from one 4 x 8 ft sheet (a three-stage cut: the bridges and struts come out of the strips the walls leave), and 1x2 cleats, with every wall 25 mm above the pour line, held by printed brackets and saddles with M4 bolts through that band, so no bolt passes through a cast face. Every printed piece bolts to its board from outside into a captive nut (a bayonet pocket in the face against the board), or pins on if it is too thin, and pieces cut to fit the printer are pegged together. The hidden pocket cores are XPS foam, which keeps the printing to about 1.8 kg of PETG, or 1.1 kg with the base's bevel strips from a 2x3 (sliced in Bambu Studio). A clash check in the model keeps the kit, the boards, the castings and the steel apart. Two joints are made to fit by construction instead of by measuring: the front base half is cast against the rear half (flush top faces, matching split faces, end plates straight onto the dowel pins), and the dowels are grouted into oversize holes at first assembly, so two independent molds don't have to agree. The two uprights are the same part, so one mold casts both, one after the other.

- The mix matters as much as the geometry. Use a well-graded aggregate (gravel up to about 10 mm, sand, fine sand, and whiting, i.e. calcium carbonate powder) and vibrate it. The split in the build guide follows a modified Fuller curve, P(d) = (d^q - d_min^q) / (D^q - d_min^q) with D = 10 mm and q about 0.45: q = 0.5 (classic Fuller) is a little coarse and needs more epoxy to wet, q = 0.37 (Andreasen) packs tighter but its extra powder soaks up epoxy.

- **Aluminum oxide in the beam and uprights.** Alumina grit (E about 370 GPa for brown fused, a little less for black, against about 70 for quartz and granite) replaces the coarse sand, volume for volume, in those three castings only (`analysis/alumina.py`): brown #16, one 50 lb bag. A differential-effective-medium estimate of the mix (the plain mix comes out at 34 GPa, close to the 30 GPa the frame model uses) puts it 22 % stiffer with the #16 alone, 10 % with black 70 grit alone (for the fine sand) and 37 % with both. On the frame with 3/8 in beam bars, where the steel skeleton carries most of the load, that is worth much less than it was on the all-epoxy-granite frame (+17 %):

Grit in the beam and uprights (3/8 in beam bars)	X	Y	Z	Grit
None (sand)	89.2	137.8	117.0	
#16 in the beam only	92.8	147.5	124.1	~$70 (two 12 lb bags)
Both grits, beam only	95.0	153.1	128.4	~$105
#16, beam and uprights	96.5	150.2	128.8	$110
Both grits, beam and uprights	101.0	157.8	136.3	$150

  So the #16 is about +9 % for $110 and the 70 grit another +5 % for $40. - **Steel instead of the grit** (`analysis/steel_study.py`). Steel is about 200 GPa and costs about $2/lb as flat bar, so the same money buys more stiffness as thicker skeleton parts than as grit, if it goes where the frame bends. Each part was thickened in turn on the sand-only frame (its steel scaled up where it lies) or taken out (filled with epoxy granite):

Plain sand, 3/8 in beam bars, and ...	Steel	X	Y	Z
nothing changed		89.2	137.8	117.0
beam face and back bars 1/2 in	+3.4 kg	92.7	142.3	122.6
beam bars 5/8 in	+6.8 kg	95.6	145.9	127.1
upright columns 3/16 in wall	+1.5 kg	90.4	139.8	120.0
girders 3/16 in wall	+2.8 kg	90.0	139.2	118.2
top plates 5/8 in bars, feet 1/2 in	+2.1 kg	89.9	138.5	118.3
rail bars, seats and pedestal bars 1/2 in	+3.7 kg	89.5	138.3	117.3
beam seat and cap bars 1/2 in	+0.8 kg	89.5	138.3	117.6
beam bars 3/4 in, every 2x2 column 3/16 in wall	+12.6 kg	100.2	151.2	134.6

  (With the #16 grit, taking the beam's back bar out costs 2-3 % and the base pedestal bars under 1 %, but both are links in the clamping chain, so they stay.) Nearly all the return is in the beam bars and the columns: the beam is where the frame bends. So the frame now has 3/4 x 5 in beam bars (2-1/2 in pipe nipples between them, so the back bar keeps its cover) and 2 x 2 x 3/16 in tube for every column, about 27 lb more steel ($60), and plain sand in every casting. Voxelized as drawn that is **100.9 / 155.4 / 135.6 N/um**, +5 / +3 / +5 % over the #16 grit frame, for about $45 less; the castings are 8 kg lighter (the grit is heavy) and the frame about 4 kg heavier overall. The #16 grit on top is still an option (decided before the pour): 108.0 / 166.9 / 147.3 N/um for $110. It's an estimate, so the build guide's tap-tested bars check the mix either way. (That was before the moving-parts study below showed

the frame is only 2-5 % of the deflection at the tool. The lean-frame study took most of this steel back out for under 0.5 % there: below.) Other options: **Graded industrial quartz** (Agsco and similar) is clean, dry and consistent but costs more and ships heavy; pool filter sand is the local equivalent. **Ceramic microspheres** (Zeeospheres) help the mix flow at low epoxy, but whiting does most of that for far less. **Epoxy grout** (Laticrete SpectraLock) is fine for a test bar or a repair, but it's fine-grained (about 15-20 % binder) and far too expensive for 130 kg. Whiting is used instead of silica flour because it has no crystalline silica: a plain dust mask is enough. A good mix gets to 8.5 % epoxy, which fits a single 3 gal kit (10 % needs a second, small kit); the build guide's jar test and test bars check it before anything is cast.

What the rails and supports bolt to. Steel. The rails screw into steel risers, and the risers are bedded on the castings with a thin epoxy grout and held down by counterbored M5 screws (under the rail, between its own screws) into the skeleton: the base's tapped rail bars (Y) and the beam's tapped face bar (X). The Y screw's BF12, and the HM12-57 bracket that is its fixed end and motor mount in one (the kit's BK12 is a spare), bolt into tapped pads on the base grid's cross tubes; the bracket's pad is 3/8 x 3-1/2 in bar, wide enough for its six holes. The X supports (BK10/BF10) bolt into the beam's face bar; X and Z get HM10-57 brackets too, drawn once the one on hand is measured (no drawing is published). Behind every tapped hole there's room (the casting is relieved where the greased mold screw stood, or a relief hole is drilled in the tube below). The Z axis mounts on the X carriage and the Z plate, both steel, ground flat and parallel with the Z risers (the rails and blocks sit on them metal to metal, with no epoxy bed under them).

Load path. A cutting force goes spindle, Z plate, Z and X blocks, X rails, steel risers, the beam's face bar and the steel behind it; at each end the seat bar bears on the upright's top plate, clamped by four 3/8-16 studs; down the upright's columns to its foot bar, which bears on the base pedestal's seat bar; the three tie rods clamp that whole stack (top plate, columns, foot bar, seat bar, pedestal spacers, bottom bar) from the top plate to the nuts under the base; across the base grid (the cross tubes and their rods, the girders, the split end plates on their dowel pins, clamped by the draw rods); and back up through the Y rail bars, risers, rails and blocks to the table. Steel all the way, and every nut in that chain can be reached and retightened: the tie-rod nuts under the base and in the toe pockets, the draw-rod nuts in the front and back pockets, the stud nuts on the beam top. The epoxy granite fills round and between the steel, in compression where the rods clamp; it damps the frame and adds stiffness. The stiffness study treats it as bonded to the steel.

The steel is bolted, not welded. A welded skeleton would have to be welded before casting (and pulls out of square doing it) or after (the heat ruins the epoxy next to it), and it would make the frame one piece. Bolted from stock and drilled from templates, it goes together without a jig, the mold screws and dowels put every tapped face where the templates say, and the five castings still come apart.

Steel tolerances. A supplier's saw cut is good to about 1/8 in, and A500 tube is about 0.5 mm off nominal with a weld seam inside. Each piece is set from one end and its other end has room (`tube_cut_tol`, checked in the report). Each upright column is clamped up against its top plate, its foot in a 4.8 mm epoxy bed on the foot bar; each girder is clamped between its end plates by the draw rod, set from the split; the pedestal spacers set where the bottom bar sits, which only moves it a little against the underside; the rods are screwed up to a stop at one end and have a 6 mm tail past the far nut. The printed rings that center the columns on their rods have 0.5 mm a side (`tube_fit`) and grooves for the seam. If the tube that arrives is further off, put its calipered sizes in `frame_tube` / `post_tube` and re-run before printing.

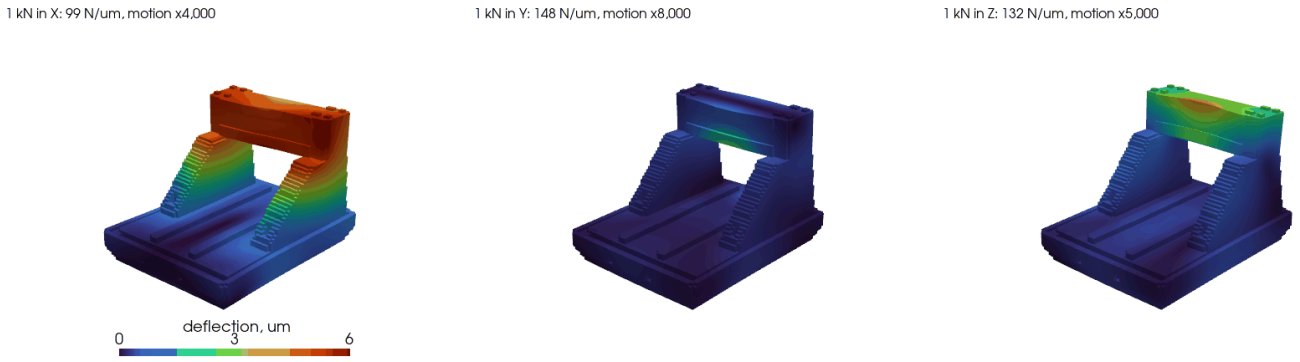
The enclosure's 2020 frame bolts to the base top with angle brackets into cast-in M5 inserts, so the base is the enclosure floor. Seal the top with an epoxy gel coat; coolant runs forward over the graded floor and along the side gutters to the trough and its one drain at the front. The door laps over the base front face.

Assembly: base halves (end plates onto the dowel pins, tighten the draw rods), uprights (dowels grouted into the feet the first time, tighten the tie rods), beam (cap screws), axes, enclosure. To break it down, take the X and Y motors off first (they pass through the enclosure panels), then reverse the order. Heaviest piece: the rear base half (about 42 kg). The upright feet run forward across the base split, so the base halves have to be joined before the uprights go on.

Stiffness study (`analysis/`)

A finite-element model of the frame was used to see where the epoxy granite does the most good, and a topology optimization was run on the base.

Model. The castings and cast-in steel are voxelized from this model at 10 mm (66k brick elements). Each voxel's stiffness is its mix of epoxy granite (E = 30 GPa) and steel; the joints between castings are treated as bonded, which is what the preload is for. A 1 kN force on the tool goes into the beam through the X blocks (the X component through the X ballscrew to its BK bearing) and, reversed, into the base through the Y blocks (the Y component through the Y screw to its BK). The result is the frame's own stiffness at the tool, tool against work, for X at the center and end, the table at both ends and the center, and the tool 50 mm above the table. Rails, blocks, screws and the spindle are not in it: those are the moving parts' study below, which puts the frame in series with them.



In X the beam is pushed sideways and the uprights lean with it, so the uprights and base take a good part of the X deflection; in Y and Z it is mostly the beam bending between the uprights.

Findings (frame only, N/um at the tool, mean over the load cases):

Frame	Epoxy granite	X	Y	Z
Previous design	42.4 L	48.9	80.7	57.9
Same, base made rigid		87.2	130.9	80.2
Same, uprights rigid		83.3	103.9	102.6
Same, beam rigid		122.9	175.7	125.9
Base with no pockets at all (+14 L)	56.8 L	61.5	90.1	67.0
Beam 120 -> 140 mm deep only (+1.5 L)	43.9 L	51.1	90.9	63.8
Optimized base + 140 mm beam	43.7 L	54.0	94.5	68.0
+ riser hold-downs, support pads, uncut 600 mm Y rails	43.7 L	55.7	96.8	68.9
Base + uprights optimized together, same material (free-form uprights)	43.7 L	63.9	103.3	85.2
Raked-buttress uprights, base re-laid around them	52.6 L	77.5	107.5	93.0

Frame	Epoxy granite	X	Y	Z
+ blunt upright toe, rounded edges, deeper slope pockets	53.0 L	78.2	110.1	95.5
+ alumina grit in the beam and uprights (estimate)	53.0 L	91.3	127.3	115.7
Lighter base (pockets under the upright feet, 20 mm skin), 9 kg less	49.5 L	84.3	124.5	111.1
Same, tie rods not counted as glued in (they sit in clearance holes; the rows below do the same; `vox_before_chain`)	49.3 L	81.6	121.9	107.6
+ tie rods right through the base to steel anchor bars, two-layer top plates	50.5 L	86.8	124.7	111.6
+ posts and sockets taken out, a second dowel per foot	52.4 L	86.3	124.2	110.1
+ upright feet 260 mm ahead of the beam (from 290)	51.3 L	84.2	123.0	109.3
+ steel skeletons in the uprights and a steel box in the beam (all 3/8 in)	47.9 L	96.0	132.7	130.9
+ steel grids in the base (all 3/8 in)	45.8 L	107.1	153.5	138.1
Slimmed: webs, cross plates and braces 1/4 in with windows, 3/8 in foot plates (64 laser-cut plates)	48.5 L	99.7	146.9	124.7
All stock flat bar and tube, bolted (no custom parts), beam 146 mm high	44.8 L	101.0	157.8	136.3
The same with the #16 grit only (the 70 grit an option)	44.8 L	96.5	150.2	128.8
Plain sand, the grit's money in steel: 3/4 in beam bars, 3/16 in column walls	44.9 L	100.9	155.4	135.6
The base's lower edge beveled 1-1/2 x 2-1/2 in, the legs tucked in behind it	44.6 L	103.4	155.8	136.5
+ the uprights' backs raked, each foot 40 mm deeper	45.4 L	105.2	157.2	138.2
Stock flat-bar plates (a 6 in X carriage, so the uprights are 18 mm closer and the beam shorter), the HM12-57 on Y	44.1 L	107.4	168.2	146.4
Now: 1/2 in face and 3/8 in back beam bars on 3 in nipples, 1/8 in column walls (the lean frame, below)	**44.8 L**	**98.3**	**153.1**	**131.3**

- **Steel skeletons.** Making the steel the frame (epoxy granite cast round it) makes the frame stiffer than the last all-epoxy-granite design. The first version was 64 custom laser-cut plates slotted together (+18 / +19 / +14 %); the one drawn now is all stock flat bar and tube, cut to length, drilled from templates and bolted, and comes out stiffer still: +20 / +28 / +25 % (101.0 / 157.8 / 136.3 N/um with both grits), with modes 132 / 262 / 299 Hz, about 60 kg of steel and hardware and 113 kg of castings (the hollow girder tubes take some epoxy granite out of the base). With the grit's money spent on steel instead (below) it's 100.9 / 155.4 / 135.6 N/um in plain sand, 73 kg of steel and 105 kg of castings. It loses the laser version's upright brace and beam diaphragms but gains the base girders' and the beam bars' depth.

- **The steel chain** (tie rods from the top plates to anchor bars under the base) is worth +6 / +2 / +4 % (the bars, the thicker top plates and the solid band left under each foot together). It also makes the posts redundant: with the feet clamped through steel they were worth under 1.5 %, so they're gone, and with them the sockets, the post bushings and the post grouting (a second grouted dowel per foot locates it instead). The stiffness left over paid for 30 mm off each upright foot. The frame ends up +3 / +1 / +2 % stiffer than before the chain, with modes 136 / 268 / 303 Hz (from 132 / 265 / 282), and about 6 kg heavier (the bars and the thicker top plates). A 250 mm foot keeps the static stiffness but drops mode 2 below the old frame; shorter than that the front foot pockets run past their mold bridges. Thinner uprights and a thinner front wall didn't pay.

- The beam is the biggest share of the deflection, then the base and the uprights about equally. Liter for liter, epoxy in the beam is worth roughly ten times as much as epoxy in the base, so the beam now fills the upright tops (140 deep instead of 120).

- In the new frame the beam is still the biggest share. Making the uprights rigid would add about 75 / 25 / 85 % in X / Y / Z, and making the base rigid about 70 / 70 / 35 %. The uprights are only 3.1 L each, so per liter they're the next place to add material (wider or deeper, which also grows the base and the enclosure).
- **Uprights.** Optimizing the base and the uprights together (`analysis/topopt_frame.py`, one shared budget, the uprights free to grow anywhere beside the table under the beam) moves about 3 L from the base into the uprights and turns each upright into a raked buttress: the foot runs about 300 mm forward along the base and the front face slopes up to just ahead of the beam (`analysis/figs/frame_34.6_upright.png`). That alone, at the same total material, is +15 / +7 / +24 % in X / Y / Z. With 4 L less material in total the frame ends up about 20 % less stiff than before, so cutting material isn't worth it.
- The castable version drawn now is a solid full-width buttress (one sloped board in the mold, no cores) with its foot 290 mm ahead of the beam face: 5.3 L per upright (5.5 L since the toe got a 20 mm nose). The base was then re-optimized around the longer feet: it wants full depth under the feet and a wider tie across the split. Total +8.9 L of epoxy granite (about $75) for +39 / +11 / +35 %.
- The base optimization (`analysis/topopt.py`, pockets open to the bottom) puts the base material into girders under the rails, a tie between them behind the split and knees to the sockets, and takes it out of the perimeter walls, the cross rib at the split and the skin. The castable version drawn here has the same volume as before (28.4 L) and is 5 % stiffer on average (at least 3 % in every load case); the beam change adds the rest.
- The cast-in steel is mostly there for the joints. Replacing the base tubes with solid casting would cost only about 1 %; the beam tubes are worth 6-7 %. Six #6 rebars instead of the beam tubes would be about 3 % less stiff, so the tubes stay.
- Cutting epoxy from the base is cheap in money but expensive in stiffness. At the optimized layout's best: 24.5 L matches the old base, 21 L loses 11 %, and below about 18.5 L (the minimum skin, walls and cover around the steel) it falls off fast. Epoxy granite costs about $8.50/L, so even the best case saves about $35. A free-form base with closed box sections (lost foam cores) is only about 8 % stiffer than open pockets at the same volume, which isn't worth the trouble.
- **Weight** (`analysis/weight.py`, `to_light_*`). With the alumina grit in the beam and uprights there was stiffness to spare, so the base was optimized again at smaller budgets. Free-form (ideal) pockets: 27 L of base is 14 kg lighter for -7 / -1 / -2 %, 24 L is 21 kg lighter and still at the sand-only frame's stiffness, 21 L drops below it. Nearly all of it comes out from under the upright feet, which the earlier, larger budget had filled. The castable version keeps a rib across the front of each foot and along the split and leaves room around the tie-rod couplers and the sockets: base 32.8 -> 29.1 L, castings 9 kg lighter, frame -8 / -2 / -4 % against the alumina frame and still +8 / +13 / +16 % over the sand-only one. The simple trims don't pay: 60 mm girders leave under 5 mm of casting beside the rail tubes, and a 55 mm front wall runs the pockets into the legs.

Material and the 8.5 % mix. The model uses E = 30 GPa for the epoxy granite. Published values for well-graded 7-10 % mixes are 25-45 GPa (one commercial mineral casting lists 25 GPa); the university studies use 15-30 % resin with powder fillers and aren't comparable. Granite is about 70 GPa and the epoxy about 3 GPa, so a lower epoxy fraction is as stiff or stiffer as long as every stone is wetted and there are no voids; what it gives up is a little damping (the resin does the damping) and some margin against dry spots. The frame's stiffness scales almost directly with E:

E of the epoxy granite	X	Y	Z
20 GPa	58.0	80.3	72.3
25 GPa	68.3	95.5	84.1

E of the epoxy granite	X	Y	Z
30 GPa (assumed)	**78.2**	**110.1**	**95.5**
35 GPa	87.9	124.4	106.6
40 GPa	97.3	138.4	117.4

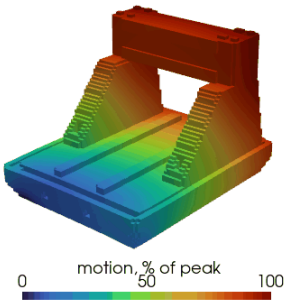
So the packing of the mix matters more than the last percent of epoxy. The build guide's jar test and tap-tested bars check both before anything is cast.

Vibration (`analysis/modal.py`). The frame on its four feet, with 22.2 kg of X carriage (steel, with its angles), Z axis (steel Z plate) and spindle on the beam blocks and 12 kg of table, vise and work on the Y blocks, as drawn now (plain sand, 1/2 and 3/8 in beam bars, the legs tucked in behind the base's bevel, the uprights' backs raked), with the optional #16 grit on top (`FEA_GRIT=1.22`), and with the legs in the corners and a square edge, as before. Tucking the legs in shortens the span the base bends over between its feet: +17 % in the first mode and +31 % in the base's own bending, more than any change to the castings has bought (the grit and corner-leg columns were run on the 600 mm wide frame, before the 6 in carriage brought the uprights in, and with the 12 kg aluminum-carriage head; the steel carriage and Z plate, about 10 kg more, take the drawn modes from 149 / 300 / 391 / 400 Hz to 145 / 271 / 371 / 376 Hz, and the lean frame's thinner steel to the figures here):

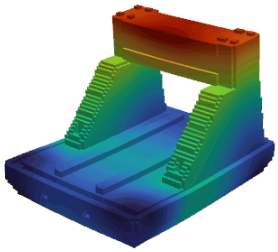
Mode	Drawn	+ #16 grit	Legs in the corners	Motion
1	**148 Hz**	151 Hz	128 Hz	gantry sways side to side (X); the base twisting on its feet carries a third of it
2	**271 Hz**	303 Hz	256 Hz	beam rocks front to back (Y)
3	**365 Hz**	391 Hz	298 Hz	base bends, front to back
4	**370 Hz**	410 Hz	349 Hz	gantry sway with the base, second form

Bricks this coarse run a little stiff: the same frame in 7.5 mm voxels comes out 3-10 % lower in its modes (and within 3 % in static stiffness), so the real frequencies are probably 10-20 % lower. All four are inside the tooth-pass band for steel (roughly 130-400 Hz: 4,000-8,000 rpm with 2-3 flutes). That's normal for a machine this size; the frame is still several times stiffer than the spindle and tool (static X/Y/Z above), and chatter on a mill like this usually comes from the spindle, Z axis and tool. If a frequency rings, change the spindle speed by 10-20 % rather than the frame. Where the feet are matters more than the castings' shape (above); a solid bench (the base twisting on its feet is in every mode) helps too.

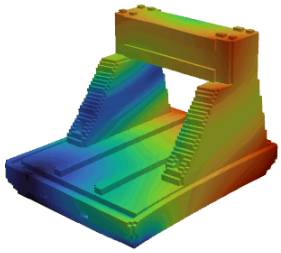
Mode 1: 148 Hz



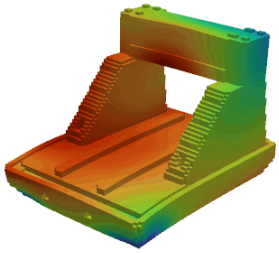
Mode 2: 271 Hz



Mode 3: 365 Hz



Mode 4: 370 Hz



Stiffness at the tool, moving parts and all (`analysis/chain.py`, `chain_geom.py`). The frame study stops at the rail blocks. This one carries on to the tool: the head (X carriage, Z plate with its risers and rails) and the table are voxel models of their own at 1/8 in (five layers through a 5/8 in plate; checked against beam theory, within 5 %, a little stiff), the rail blocks and ballscrews are springs, and the frame's compliance from the study above is added in series, load case by load case. The blocks are HIWIN's catalog rigidity for an HG15C at light preload (Z0, 196 N/um); the screws are the shaft from its fixed bearing to the nut (at the root diameter), an unpreloaded nut (about 100-150 N/um) and the bearing pair (about 80-100 N/um), the last two estimates. The spindle and its clamp are taken as rigid and the bolted joints as bonded, so the real machine is somewhat softer still.

Tool against work, N/um	**X**	**Y**	**Z**
The frame alone (above)	98.3	153.1	131.3
With the moving parts, 5/8 in aluminum X carriage and Z plate (as it was)	3.1	2.7	4.0
With the moving parts, 5/8 in steel X carriage and two angles, aluminum Z plate	3.5	3.5	6.6
With the moving parts, steel X carriage with its angles and steel Z plate, the frame before the lean steel	3.9	3.8	6.9
The same on the lean frame (drawn)	**3.8**	**3.8**	**6.8**

The moving parts are where the machine gives: the frame is 2-5 % of the deflection at the tool. With the aluminum carriage the carriage itself was the softest link, half of the Z deflection and a third of Y's: between the X blocks it bends where the Z blocks bear on it, and above them it cantilevers about 210 mm up to the Z screw's top bearing, which takes the vertical cutting load 30 mm in front of the plate. A thicker carriage moves the spindle away from the beam and gives some of the gain back; steel at the same thickness doesn't, costs less than the aluminum bar (about $35 at the yard against $66 or more new) and only weighs 5 kg more. The two 1-1/4 x 1-1/4 x 1/4 in angles bolted up its back, above the beam's drip shield, stiffen that cantilever. With the carriage fixed, the aluminum Z plate was next (a fifth of X and Y): steel there too, the same 5/8 x 6 in bar for less money, 3.7 kg more on the Z screw. What each change is worth (`python chain.py <variant>`):

Change from the drawn machine	X	Y	Z
5/8 in aluminum carriage and Z plate, no angles (as it was)	-20 %	-30 %	-42 %
5/8 in aluminum Z plate	-9 %	-9 %	-4 %
No angles	0	0	-8 %
Medium-preload blocks (ZA, 365 N/um: genuine HIWIN, a bolt-on upgrade)	+28 %	+24 %	+12 %
Blocks at half the catalog rigidity (a loose clone: check them for play)	-31 %	-27 %	-17 %
Y screw fixed at both ends (the spare BK12 for the BF12)	0	+5 %	0
Screw bearings or nuts twice as stiff	+1-3 %		
1 in table, 1 in or 3/4 in Z plate	+3-9 %		

(The last two rows were run with the aluminum Z plate.) Drawn, the deflection is shared by the Z blocks and Z plate (45 % of X and 35 % of Y), the X blocks (12-18 %), the table and the Z screw (a quarter and a fifth of Z) and the carriage (20 % of Y and 22 % of Z). That is the order for any later upgrades; the frame doesn't need anything. It also says the frame has stiffness to spare, which is right for the one part that can't be changed after the pour: everything that bolts on can be upgraded later.

The lean frame (2026-10-27) (`analysis/lean_study.py`, `topopt_gantry.py`). If the frame is 2-4 % of the deflection at the tool, some of its steel and casting is buying very little. Each change was voxelized (or the steel scaled where it lies, as in the steel study), judged at the tool with the moving parts in series, and by its modes, since those are what thinner castings give up first:

Change from the frame before (10 mm voxels)	At the tool X / Y / Z	Modes, Hz	Castings	Steel	Cost
Before	3.86 / 3.81 / 6.86 N/um	145 / 271 / 371 / 376	103.5 kg	75.1 kg	
Beam bars 1/2 in	-0.3 / -0.1 / -0.4 %	148 / 276 / ...		-5.9 kg	-$26
Beam face bar 1/2 in, back bar 3/8 in	-0.3 / -0.2 / -0.5 %			-7.4 kg	-$32
2x2 columns 1/8 in wall	-0.1 %			-3.1 kg	-$20
Top plates of 3/8 in bars	about 0			-1.2 kg	-$5
Drawn: the beam bars 1/2 and 3/8 in (3 in nipples), the columns 1/8 in wall	**-0.3 / -0.2 / -0.5 %**	**148 / 271 / 365 / 370**	**105.4 kg**	**64.5 kg**	**-$50**
... and the uprights' toes 200 mm ahead of the beam instead of 260	-0.6 / -0.3 / -0.7 %	146 / 240 / 361 / 369	101.3 kg	62.9 kg	-$3 epoxy
... toes 150 mm ahead	-0.9 / -0.4 / -1.1 %	145 / 219 / 354 / 370	96.5 kg	62.9 kg	-$10 epoxy
... the base 100 mm high instead of 110	-0.5 / -0.3 / -0.7 %	148 / 259 / 369 / 376	97.4 kg	64.3 kg	-$8 epoxy
Uprights and beam optimized together, 70 % of their epoxy granite kept (on the frame before)	-0.7 / -0.2 / -0.8 %	144 / 234 / 348 / 375	91.3 kg		-$17 epoxy

Change from the frame before (10 mm voxels)	At the tool X / Y / Z	Modes, Hz	Castings	Steel	Cost
The same, 60 % kept	-1.5 / -0.8 / -2.4 %	145 / 188 / 330 / 372	87.2 kg		-$22 epoxy

The thinner steel is nearly free: 23 lb and $50 less for half a percent at the tool, the modes the same. (The top plates stay 1/2 in bars: the lower one holds the 1/2-13 tie rods' and the beam studs' threads, and 3/8 in would leave them three quarters of a diameter of thread. The face bar stays 1/2 in for the M5 and M6 threads of the rails and supports, over two diameters.) The 3 in nipples keep the back bar where the 3/4 in one was, so the beam's depth between its bars doesn't change.

The castings are another matter. The optimizer (uprights and beam on one budget, free in 3D, mirrored, kept solid 10 mm round the steel, 15 mm round the tie rods and in a 20 mm skin on the beam) takes the material out of the uprights' raked toes and the beam's core between its bars, between the nipples (below), and the second mode, the beam rocking front to back, pays for it: 271 Hz to 234 Hz at 70 %, 188 Hz at 60 %. Shortening the toes outright is worse still per kilogram. And epoxy granite saves money only in kit steps: the 3 gal kit stays until the castings are under about 86 kg, which none of these reach without giving up a fifth of that mode or more. So the castings stay as they are, and the money is in the steel.

Is 10 mm fine enough to trust? The optimization was repeated on the lean frame at 7.5 mm (with the half model, below). It takes out the same places: the toes, the thin raked tail behind each upright, and the beam's core. 77 % of what the 7.5 mm run removes the 10 mm run removes too, and the details differ. Judged the same way, the two come out alike:

Keep 70 % of the uprights and beam (lean frame)	At the tool X / Y / Z	Modes, Hz	Castings
10 mm voxels	-0.8 / -0.3 / -0.8 %	154 / 242 / 344 / 362 (from 154 / 271 / 361 / 370)	92.8 kg (from 105.5)
7.5 mm voxels	-0.9 / -0.4 / -1.2 %	150 / 244 / 331 / 344 (from 149 / 269 / 343 / 353)	93.2 kg (from 106.1)

The frame's own stiffness is another matter. 8-node bricks with the steel and epoxy granite mixed in each voxel converge slowly, and the thin steel (1/8 in tube walls) is smeared over the coarse voxels:

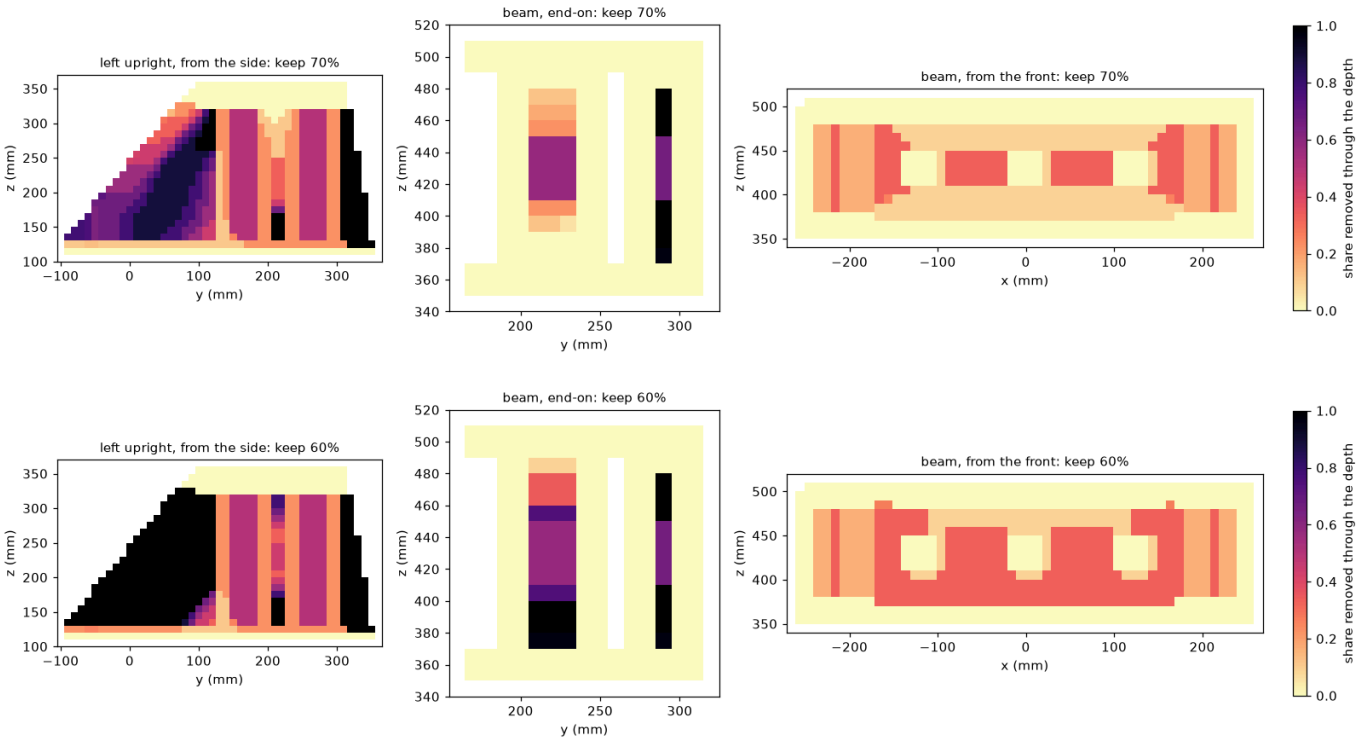
Lean frame, half model	X	Y	Z
10 mm voxels	98.7	154.4	133.4
7.5 mm voxels	95.4	151.0	125.5
5 mm voxels (1.7 million, 9.7 GB)	85.5	132.5	113.6

So the frame is probably 15 % or more softer than the 10 mm figures in these notes, and the modes lower too. At the tool that is under 1 %, since the frame is 2-5 % of the deflection there: the comparisons and the decisions hold, but the absolute frame figures are optimistic.

Which figure is right was checked with 10-node tetrahedra (`analysis/tet_check.py`, cad-rust-app's FEA). Each part is meshed at its own size and tied to the next where they touch, so the tube walls get their own fine elements. The model is one upright with its steel, cut to the slab every voxel grid shares (z = 120 to 330 mm), its bottom section held and 1 kN spread over its top section:

Upright slab, N/um	X	Y	Z
10 mm voxels	153.9	1,024	3,793
7.5 mm voxels	145.3	933	3,515
5 mm voxels	140.7	918	3,559
tetrahedra, 16 mm casting / 3.2 mm tube / 6.4 mm bar (133,000 elements)	143.3	929	3,428
tetrahedra, 12 / 2.5 / 4.8 mm (425,000 elements)	142.3	923	3,483

The two tet meshes agree within 2 %, and the voxels close in on them. At 10 mm the voxels are 8-11 % too stiff; at 7.5 and 5 mm they are within 2 %. So the 5 mm frame figures (85.5 / 132.5 / 113.6 N/um) are the ones to believe, and the 10 mm ones in these notes are about 13-15 % high.



Lightening holes in the uprights (`analysis/topopt_upright.py`, `compare_uprights.py`). The uprights were optimized again on their own, with the cuts forced to go straight through (windows) or to stop at a 20 mm web on the inner face (pockets from the outside), keeping 90 % and 80 % of their material (run on the frame before the toe and edge cleanup, which added 1-3 %):

Uprights (both)	Removed	X	Y	Z	Mode 1 / 2	Epoxy saved
Solid, as drawn		77.5	107.5	93.0	135 / 264 Hz	
One window each, 90 %	1.05 L, 2.5 kg	75.0	106.5	91.2	136 / 263 Hz	~$3
One outer pocket each, 90 %	1.05 L	75.6	106.8	92.0	136 / 261 Hz	~$3
Two windows each, 80 %	2.1 L, 4.9 kg	71.7	104.8	88.9	136 / 255 Hz	~$6
Two outer pockets each, 80 %	2.1 L	73.1	105.6	90.3	135 / 254 Hz	~$6

The holes land where you'd expect, in the low-stress middle of the buttress just ahead of the front post, but they don't pay: 10 % of the upright costs 1-3 % of the frame's stiffness and saves about $3 of epoxy (the 3 gal kit is bought either way), and the natural frequencies don't move because the mass that matters is the beam and the head on top. Each window would also need a printed liner. The uprights stay solid; the frame optimization above already moved material into them, not out.

Vibration and damping (2026-10-28) (`analysis/damping.py`, `steel_opt.py`, `topopt_damped.py`, `ring_check.py`). Static stiffness says how far the tool is pushed; what makes chatter is how far it moves at a resonance, which depends on damping as well. Each material gets a loss factor, epoxy granite 0.02 and steel 0.0005 (env `FEA_ETA_EG`, `FEA_ETA_ST`). Each mode's loss is the strain-energy-weighted mix of the two, and the peak at the tool is that mode's share of the cutting force over its damped stiffness (the frequency response by modal superposition, checked on a cantilever: peak 97.0 times static against 0.97/η = 97). On the lean frame (10 mm voxels, half model):

Mode	Hz	Loss	Steel's share of the strain	Worst peak at the tool, N/um
1	153	0.015	25 %	9.1 (X)
4	369	0.013	35 %	3.3 (X)
6	532	0.012	41 %	2.3 (Z)
7-16, up to 1.7 kHz		0.012-0.015	24-39 %	8.0 and up

Tool against work, frame alone	X	Y	Z
Static (weakest case)	94.8	140.4	124.1
At the worst resonance	3.3	3.3	2.3
Amplification	29x	42x	54x

The castings damp: every frame mode has a loss of 0.012-0.015, 25-30 times bare steel's, because 60-75 % of the strain is in the epoxy granite. Even so, at its worst resonance the frame drops to 2-3 N/um, the same as the whole machine statically (3.8 / 3.8 / 6.8 N/um at the tool), so those few frequencies are the ones to keep the tooth-pass rate off (change the spindle speed 10-20 %, as above).

The optimization ran in two stages. Stage 1, the steel, for stiffness and cost (`steel_opt.py`): each group of stock is moved one catalog size at a time, at about $2/lb for flat bar and $3/lb for tube, judged at the tool with the moving parts in series:

Steel, one size at a time	At the tool	Steel	Cost
Upright feet 3/8 to 1/4 in	-0.00 %	-1.0 kg	-$4
... seat and bottom bars 3/8 to 1/4 in	-0.01 %	-3.3 kg	-$14
... beam back bar 3/8 to 1/4 in	-0.05 %	-4.9 kg	-$21
Beam face bar 1/2 to 3/4 in, columns to 1/4 in wall, seat and cap bars 1/2 in, cross tubes 3/16 in, back bar 3/4 in	+0.42 %	+14.1 kg	+$74

Everything else is at its smallest stock size already. Nothing in the steel is worth buying stiffness with. The three thinner sizes save $21 for nothing measurable (not yet drawn: the seat and bottom bars also take the tie rods' nuts and the feet the dowels, so their threads and bearing need checking first).

Stage 2, the castings, for damping and cost with a little stiffness (`topopt_damped.py`): the objective is a quarter the mean static compliance at the tool and three quarters the worst resonant peak (an 8-norm over 24 modes and the three directions). Its sensitivities come from the modal expansion with the higher modes' static flexibility added, checked against finite differences within 1.3 %:

Uprights and beam (lean frame)	Worst peak X / Y / Z, N/um	Static at the tool	Castings
Drawn, solid	3.32 / 3.31 / 2.36		105.5 kg
Optimized for static stiffness, 70 % kept	2.61 / 2.87 / 2.26	-0.8 / -0.3 / -0.8 %	92.8 kg
Optimized for damping, 70 % kept	2.74 / 2.56 / 2.42	-0.7 / -0.8 / -1.4 %	92.8 kg
Optimized for damping, 85 % kept	3.10 / 3.12 / 2.44	-0.3 / -0.2 / -0.3 %	99.1 kg

The shape of the castings barely moves the damping: whatever is taken out, the modes keep a loss of about 0.013-0.014, and the worst peak follows the mass and stiffness given up. So the castings stay solid. The levers for damping are elsewhere: the mix's own loss factor (more resin damps more), the bolted joints (not modeled, and they damp a good deal), and dampers where something actually rings.

What rings. Thin steel can ring like a tuning fork at frequencies the frame model can't see, with the little damping steel has. `ring_check.py` meshes each piece in 10-node tetrahedra with the casting round it (clamped where the model is cut) and lists every mode under 2 kHz whose loss is under 0.005:

Piece	Modes under 2 kHz	Loss	Verdict
A leg, its top cast into the base	none		fine
An upright column in its casting	none		fine
A base girder (2x3x1/8 tube) and its rail bar	none		fine
The 6061 table, 10 x 3/4 in, on its four Y blocks	1,781 and 1,820 Hz, the plate bending	0.0003 (x 3,000 at resonance)	**rings**
The X carriage with the Z axis and spindle; the X rails on the beam face	not yet run		

The tubes are boxed in by the castings and don't ring even hollow; they're filled with dry sand anyway (above), which damps their walls at any frequency, above 2 kHz too. The sand's mass is left out of the frame model (11 kg, mostly in the base on its feet, 1 kg in each upright). The table is bare aluminum and rings: its two plate modes sit near the upper harmonics of the tooth-pass band (24,000 rpm with 3 flutes is 1.2 kHz). A vise or fixture clamped across its middle stiffens and damps it; failing that, a constrained-layer damper on its underside between the blocks.

Figures in `analysis/figs`: `frf_vox_sym10.png` (the frequency response at the tool) and `damped_vs_static.png` (where the two optimizations take the casting out).

Figures in `analysis/figs` (and `renders/frame_base_underside.png`): `frame_34.6_upright.png` (base and uprights optimized together), `base_current.png` (the previous base), `base_final.png` (solid depth from the top in each column), `tradeoff.png` (base stiffness against base volume), and the animations `deflection.gif` (the frame under the cutting loads) and `modes.gif` (the four lowest vibration modes).

Rerun it in the `diycnc` conda env (`environment.yml`: numpy, scipy, matplotlib, scikit-sparse 0.5 or later, and pyvista and imageio for the animations; older scikit-sparse falls back to a much slower solver):

```
conda env create -f environment.yml                # once (and build cadrust into it, see
environment.yml)
conda activate diycnc
python analysis/voxelize.py                        # -> analysis/vox_current.*
cd analysis
python baseline.py                                 # stiffness and each module's share (about 1
min)
FEA_VOX=vox_previous sh run_sweep.sh               # base optimizations (about 10 min each)
FEA_VOX=vox_previous python evaluate.py to_*.npz    # table + figures
FEA_VOX=vox_current python whatif.py               # stiffness of the model as drawn
# voxelize with VOX_SKIP=TieRod,DrawRod,CrossRod so the rods in their clearance holes don't count
as glued in
python alumina.py                                  # alumina grit: mix modulus and frame stiffness
FEA_VOX=vox_grit python steel_study.py             # thicker steel against the grit (about 4 min a
variant)
FEA_GRIT=1.37 python -c "import topopt; topopt.main('pocket', 24, tag='light_24')"   # lighter base
python weight.py vox_current [vox_...]             # castings mass against stiffness
python modal.py                                    # natural frequencies (about 30 s)
python animate.py                                  # figs/deflection.gif and figs/modes.gif (about
4 min)
python topopt_upright.py 0.9 [web]                 # upright holes (about 10 min each)
python compare_uprights.py to_upright_*.npz        # their stiffness and frequencies
python lean_study.py lean0: lean0:beam1238+cols18    # thinner steel and smaller castings, at the
tool (3 min each)
python topopt_gantry.py 0.7 [voxels]               # uprights and beam optimized together, 70 %
kept
# half the frame: voxelize with VOX_SYM=1 (and VOX_H=7.5 or 5 for finer voxels), then FEA_SYM=1 on
any of these
FEA_SYM=1 FEA_VOX=vox_sym5 python whatif.py
python tet_check.py                                # one upright: tetrahedra against the voxels
(10 min)
# vibration: each mode's damping and the resonant peaks at the tool (vibration and damping, below)
FEA_SYM=1 FEA_VOX=vox_sym10 python damping.py --frf # modes, loss factors, worst peaks,
figs/frf_<voxels>.png (1 min)
FEA_SYM=1 FEA_VOX=vox_sym10 python steel_opt.py    # stage 1: the steel's stock sizes, stiffness
against cost (40 min)
FEA_SYM=1 python topopt_damped.py 0.85 sym10       # stage 2: the castings placed for damping (30
min; --check, --resume)
python plot_gantry.py --ref sym10 gantry_0.7_sym10 damped_0.85_sym10   # where each one takes
casting out
python ring_check.py [legs columns girders table carriage "x rails"]    # local modes of the steel,
tetrahedra (15 min)
```

`vox_current` is whatever `build_cnc.py` draws now (`voxelize.py` rewrites it); `vox_previous` is the frame before these changes, which the optimizations and the "previous design" numbers above use. `VOX_P='{"beam_d": 160}' VOX_OUT=analysis/vox_test` on the voxelize command tries a parameter change.

Half the frame (`FEA_SYM=1`, `vfe.py`). The frame is symmetric about x = 0, so the voxels can be mirrored from their right half (voxelize with `VOX_SYM=1`, which centers the grid). Every solve then becomes two of about half the size: one for the fields symmetric about the center plane (the plane doesn't move in x) and one for the antisymmetric ones (it doesn't move in y or z), each the whole stiffness projected onto them. That is exact for any load: on the same voxels it matches the whole-frame solve to 1e-13, stiffness and modes. The supports have to be symmetric too, so the front right leg is held in x as well as the front left; the first mode comes out 4 % higher than with the one leg. At 7.5 mm it is about 30 % faster and peaks at 5.2 GB instead of 6.2 GB. Most of the memory is the assembly bookkeeping, not the factor, but it is enough to make 5 mm fit in 16 GB (about 14 minutes a solve).

Chips and coolant

Stage	What	Where in the model
Collect	Sealed base top. Inside the side gutters the whole floor falls forward to the trough, about 1:130 (1 mm below the datum at the back, 6 mm at the trough), so coolant under and behind the table runs off instead of pooling; the rail bars, screw-support pads, upright feet and inserts stand proud of it on 6 mm lands (checked in the frame study: about 1 % off the frame's stiffness, under 0.1 % at the tool, 1.5 kg less casting). Gutters between the uprights and the enclosure walls fall toward a trough across the front, which falls to one outlet	`BaseFrontCasting`, `BaseRearCasting`, `base_top_formers`
Protect	Roll-up 0.1 mm stainless strips in front of and behind the table, a spiral spring cover on the rear screw, a drip shield over the beam rails, brush seals where the motors pass through the panels	`WayCovers`, `CoverRoller*`, `DripShield`, `BrushSeal*`
Drain	The trough drops straight down through the front wall to a cast-in brass nipple over the chip basket	`DrainOutlet`
Clean	Slide-out sump under the base, between four 6 in steel legs cast in near the corners: a full-size 4 in stainless steam-table pan (~13 L, ~9 L working). Front to back: 1/3-size perforated pan as the chip basket, a bag of bar magnets, printed underflow baffle, 100 um screen in a printed frame, pump	`Sump*`, `Leg*`
Return	24 V brushless submersible pump (switched by the controller's M8 output; no mains in the coolant), push-to-connect at the pump, hose under the base and up the back to a bulkhead in the back panel, along the printed cable chain, a loop down to a manifold on the spindle clamp, two Loc-Line nozzles	`ReturnHose`, `Bulkhead`, `CoolantLines`, `CoolantNozzle*`

To empty or clean the sump, pop the hose off the pump and slide the pan out the front (set `sumpOut` in the `Params` sheet to see it pulled out). The legs raise the machine 117 mm and it still sits on four M12 leveling feet, now in the leg ends. Use a water-soluble coolant and keep it topped up; skim tramp oil off the clean end now and then.

Add later, fixed now. The covers, the sump and the enclosure can come later, but whatever they fix to is in the castings already, so nothing is recast:

Add-on	Fixing	Where in the model
Way-cover rollers	Front: an M5 insert behind the coolant trough for each end bracket's foot. Rear: an insert outboard of each roller, and an M5 hole tapped in the Y bracket's pad beside it for each inner foot	`roller_layout`, `CoverRoller*`
Spiral screw cover	None: it clamps between the nut housing and the Y bracket's bearing end	`CoverScrewRear`
Drip shield	The four front beam studs: its back edge goes under their washers (the studs are 190 mm, long enough for the 3 mm ACM)	`DripShield`
Cable-chain mount	Two M5 inserts in the beam top	`ChainMountFixed`

Add-on	Fixing	Where in the model
Enclosure	11 M5 inserts in the base top	`rail_bolts`
Flood coolant	The trough, gutters and cast-in drain; room for the sump between the legs	`base_top_formers`, `DrainOutlet`
Homing switches	None: Y on the Y motor bracket, X and Z on the X carriage	
Y cable chain (a 4th or 5th axis on the table)	Two M5 inserts in the base top, in the 49 mm gap between the table and the right upright, and two M5 holes tapped in the table's right edge	`ychain_inserts`
A 4th axis, a trunnion or a vise plate	The table is the interface: the M6 grid at 25 mm and two 8 mm dowel holes, 200 apart on its X center line, so a module lands square	`Table`

Cable chain. One printed chain on top of the beam carries the coolant line, Z motor, spindle power and spindle water hoses to the carriage (17 links, 38 x 21 outer, 20 mm pitch, R38). Print it in PETG, not PLA. The model shows a simplified link; for the real parts, start from a proven printable parametric chain sized to about 30 x 15 inside and print three links to check the fit before printing the rest. The table and Z axes don't need chains: the table carries no cables and Z uses a short service loop. A 4th or 5th axis on the table brings cables: a second chain of the same links, standing on edge in the gap between the table and the right upright, its fixed end on the two inserts cast in for it.

Parts that change shape with the axes. The way-cover strips, the screw cover, the cable chain and the nozzle hose are drawn for the home position. The collision check and the renders rebuild them at every pose, but jogging `posX` / `posY` / `posZ` in the GUI won't redraw them; re-run the macro to see them elsewhere.

Cost

Checked against real listings (2026-10-09/22, the sourcing list). The list is in three parts: **1, the frame** (castings, steel, rails, ballscrews) about **$1,576**, plus about **$183 of add-later niceties** (way covers, screw cover, stainless sump and chip basket, grease gun) that bolt on whenever they're bought; **2, the electronics and spindle** about **$418** plus the spindle the builder picks ($250-375 for a 1.5 kW 110 V kit, $430-600 for the 2.2 kW kit with a 120 V step-up VFD); **3, the enclosure and flood coolant** about $540 plus the coolant pump (the ACM sheet priced at $117 plus freight; a local quote should do better, `docs/1-sourcing/quote-requests.md`). The steel and aluminum are priced as remnants and the joint hardware from a hardware store; the remnant prices are estimates.

Frame cost-down (2026-10-12). The frame was $1,839 with everything in it. The changes, none of which gives up travel, rigidity to speak of, chip protection or the coolant system:

Change	Frame
Before	$1,839
Missing or low lines added: a third B7 rod, the PVA film, the quarter-round and silicone	+$47
Add-later niceties out of the frame total (they bolt to fixings cast in now, below; $183 with cheaper springs and screw cover)	-$210
Their stand-ins for the first chips: a grease syringe, a bin under the drain	+$20
#16 grit only; the 70 grit is an option (96.5 / 150.2 / 128.8 N/um against 101.0 / 157.8 / 136.3)	-$40
One melamine sheet (one upright mold for both, three-stage cut) and 1x2 cleats	-$36
Mold hardware and rods to the real counts; low-head screws by the piece	-$24
More fine sand (it replaces the 70 grit)	+$7
After	**$1,603**

Second frame cost-down (2026-10-22). Nothing taken out; the frame ends up stiffer:

Change	Frame
Before	$1,603
Whiting from a Baltimore pottery supplier, picked up ($17 for 50 lb against $42 plus shipping)	-$25
Couplers from AliExpress (about $2 each against $13-14 a pack; not needed until the axes go on)	-$19
B7 rod by the piece from Bolt Depot ($7.63 a rod against a $49 five-pack)	-$26
The #16 grit's job done by steel: 3/4 in beam bars, 3/16 in column walls (+27 lb), plain sand everywhere, 100.9 / 155.4 / 135.6 N/um against 96.5 / 150.2 / 128.8	-$110 +$60 +$7
Every rod a B7 stud bought at a stock length instead of zinc rod sawn by hand: nothing to cut, and stronger	+$20
The 3/8-16 beam studs and ties bought at length too, instead of three B7 rods	-$3
After	**$1,507**

The steel X carriage (2026-10-26). The moving-parts study (above) found the aluminum carriage the softest link between the tool and the work; steel at the same size is +13 / +30 / +65 % at the tool:

Change	Frame
Before	$1,507
The carriage in 5/8 x 6 in steel and two 1-1/4 in angles, with the skeletons' steel (about 20 lb)	+$40
The carriage out of the aluminum (its share of the drop estimate)	-$20
M6 screws for the angles	+$4
After	**$1,531**

Against aluminum bought new (about $66 for the carriage alone) it is about $30 cheaper, not dearer.

The steel Z plate and grinding (2026-10-27). The same study puts the aluminum Z plate next in line, and steel there is +10 / +9 / +4 % at the tool for less money. Steel bar as it comes isn't flat, though (hot-rolled bar can be half a millimeter out over the carriage's length), and the carriage and the Z plate carry the rails and blocks metal to metal, so both plates and the Z risers are ground:

Change	Frame
Before	$1,531
The Z plate in 5/8 x 6 in steel, with the skeletons' steel (about 13 lb)	+$25
The Z plate out of the aluminum	-$20
Blanchard grinding: the carriage, the Z plate and the Z risers, both faces	+$75
After	**$1,611**

The grinding is the only real addition, and it buys accuracy, not stiffness. Machining the plates on the machine itself (built first with the bars as they come, then facing a second pair and swapping them in) would cost about $35 of bar instead, but the result is only as flat as the machine's own travel, and custom-machined steel parts (online quotes run from tens to over a hundred dollars a part) cost more than the stock bar and the grinding.

The lean steel (2026-10-27). The frame is 2-5 % of the deflection at the tool, so the skeleton's heavier steel was buying little (the lean frame, above):

Change	Frame
Before	$1,611
The beam bars 1/2 and 3/8 in instead of 3/4 in, the 2x2 tube 1/8 in wall instead of 3/16 in (about 20 lb)	-$50
A little more epoxy granite where the steel was (0.2 kg of epoxy, inside the 3 gal kit)	0
Lean steel	$1,561
Dry sand in every tube and vinyl sleeves on the rods that come out (2026-10-29)	+$15
Now	**$1,576**

What would take it further, all on the builder's side of the list: a steel-yard quote for the cut list (about $300 at $2/lb, against the $375 estimate), AliExpress motor brackets (about $29 less, 2-4 weeks; clone dimensions vary), and car paste wax under the PVA instead of Partall ($15 less; test it on a melamine offcut). The aluminum table is the risk the other way: it is a stock flat-bar width, so any drop of 10 in bar only needs a length cut, but bought new and cut to length it is about $105 against the $45 estimate.

The estimate below was made before the listings were checked and is too low; it assumed AliExpress-level prices. It's kept here to show what each earlier cost-down change saves.

Rough estimates in USD at hobby-supplier prices (AliExpress/Amazon-type sources, late 2026). They're not quotes; check current listings. The cost-down column keeps the machine's capability: same travel, rigidity, spindle, closed-loop X/Y, flood coolant and enclosure.

Item	Before	Cost-down	Change
Epoxy + aggregate	$240-315	$200-255	Hollow rear base, 8.5 % epoxy mix (well-graded aggregate + whiting fines), bulk kits
Cast-in steel, rods, dowels, fasteners	$90-140	$90-140	
Rail risers	$60-90	$25-40	Cold-rolled flat bar bedded in epoxy and replicated, instead of MIC6 strips
MIC6 table, carriage, Z plate	$120-180	$120-180	Remnant/drop plate is often much cheaper
Rails + blocks (table, Z)	$110-160	$110-160	Beam rails owned
Ballscrew kits (table, Z)	$70-100	$70-100	Beam kit owned
X/Y closed-loop motors + drivers	$170-220	$140-180	2 Nm 23HS30-E1000: lower inductance, holds torque better at speed, 37 mm shorter
Brackets, couplers	$20-30	$20-30	One bracket owned
Spindle, VFD, clamp, cooling loop	$280-370	$280-370	
Electronics	$150-200	$150-200	
Enclosure	$220-300	$145-210	3 mm ACM walls and roof; polycarbonate only in the door
Coolant system	$105-180	$105-180	
Way covers, seals	$65-105	$65-105	
Lubrication	$40-60	$10-15	Grease gun on the blocks' grease nipples instead of a central pump
Gaskets, inserts, feet	$30-45	$30-45	

Item	Before	Cost-down	Change
Printed cable chain	$3	$3	PETG
Total	**~$1,780-2,500**	**~$1,560-2,210**	

Bigger savings mean giving something up. Options, roughly in order of impact:

- **Phase it.** Build without the enclosure and flood coolant first (mist or air blast). That saves about $250-400 up front. The base still has the gutters and mounting inserts, so both can be added later.
- **Polyester instead of epoxy** for the castings: about half the resin cost, but it shrinks more, so you rely more on the epoxy-replicated mounting surfaces.
- **1.5 kW 110 V spindle** instead of the 2.2 kW you can upgrade to 240 V later: about $40-70 cheaper, less power for steel.
- **Open-loop motors** on X/Y: about $60-80 cheaper, but you lose stall and crash detection.

Reference dimensions

Component	Source	Notes
HGR15 rail, HGH15CA block	HIWIN Linear Guideway catalog G99TE23-2203, HG series (p.38)	Ball-groove positions are not published; estimated.
SFU1204 / SFU1605 nuts, screw end machining	TBI Motion SFU sheet L010974 (via Anaheim Automation)	
BK10/BK12, BF10/BF12 supports	TBI Motion sheets L011175 (BK), L011178 (BF)	BK locknut size estimated. BK12 counterbore depth (1.5) is as printed; worth checking.
DSG16H nut housing	makerstore.cc drawing TR-BS-DSG16-00	
DSG12H nut housing (24 mm bore)	zyltech / vxb listings	Center height and holes estimated.
NEMA 23 motors	StepperOnline drawings 23HS45-4204D-E1000, 23HS30-5004D-E1000, 23HS30-3004S	
Motor bracket (X, Z kit style)	No published drawing	Sized from kit listings: motor face ~47 mm from the BK.
HM12-57 bracket (Y)	Dimensioned drawing in the ANWOK listing's images (Amazon B09S39Q1RT)	89 x 60 x 44, shaft 25 up, BK12 hole pattern plus two counterbored holes 40 from the motor end, 6001 bearings, coupler L30-42. The flange height and the inside are estimated.
D25 L30 coupler	Seller listings (5 N·m, 8x10 / 6.35x10 bores)	Hub depth estimated.
GDZ80-2.2 spindle	StepperOnline drawings GDZ80-2.2A/B	Some features scaled from the drawing (±1 mm). Body length varies by seller (213/230/240): measure yours.
80 mm cast clamp	cnconline.co.za 80 mm spindle bracket drawing	148 W x 80 D, 14 foot, bore center 50 off the foot.
ER20 nut	DIN 6499 / spindle drawing	Chinese spindles ship a 30 AF hex nut.
2020 extrusion	Misumi HFS5-2020 catalog p.2-511	Owned V-slot is OpenBuilds profile (6.25 throat).

Before casting anything, measure the parts you actually have, especially the owned ANWOK rails, blocks and SFU1204 kit. Clone parts can differ from these catalogs by a millimeter or two.